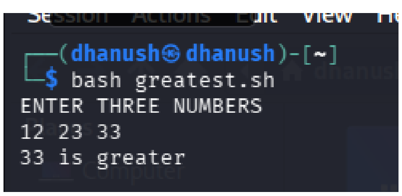
Ex. No: 2 Illustrate UNIX Commands and Shell Programming

1: Greatest Among Three Numbers

Program:

```
#!/bin/bash

echo "ENTER THREE NUMBERS"

read a b c

if [ $a -gt $b ] && [ $a -gt $c ]

then

echo "$a is greater"

elif [ $b -gt $c ]

then

echo "$b is greater"

else

echo "$c is greater"

fi
```

Output:



2. Factorial of a Given Number

Program:

```
#!/bin/bash

echo "ENTER THE NUMBER:"

read n
```

fact=1

while [$n -gt 1]

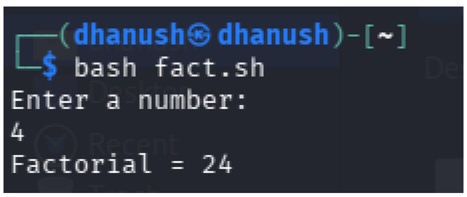
do

fact=$((fact * n))

n=$((n - 1))

done

Output:



3. Sum of Odd Numbers up to N

Program:

#!/bin/bash

echo "ENTER THE RANGE:"

read n

x=1

sum=0

while [$x -le $n]

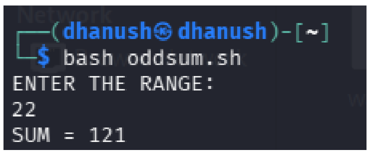
do

sum=$((sum + x))

x=$((x + 2))

done

Output:

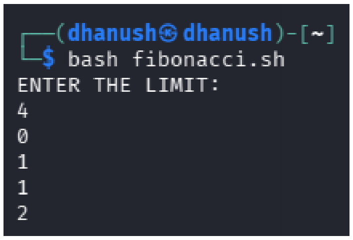


4. Generation of Fibonacci Numbers

Program:

```
#!/bin/bash

echo "ENTER THE LIMIT:"

read n

p=-1

q=1

i=1

while [ $i -le $n ]

do

r=$((p + q))

p=$q

q=$r

echo "$r"

i=$((i + 1))

done
```

Output:

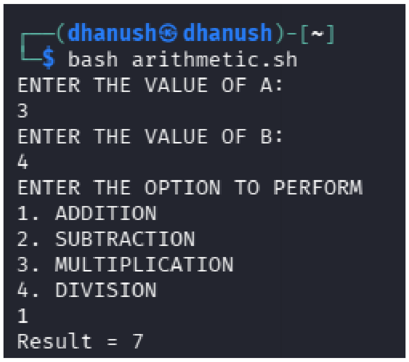


5.Arithmetic Calculator

Program:

```
#!/bin/bash

echo "ENTER THE VALUE OF A:"

read a

echo "ENTER THE VALUE OF B:"

read b

echo "ENTER THE OPTION TO PERFORM"

echo "1. ADDITION"

echo "2. SUBTRACTION"

echo "3. MULTIPLICATION"

echo "4. DIVISION"

read op

case "$op" in

1) echo "Result = $((a + b))" ;;

2) echo "Result = $((a - b))" ;;

3) echo "Result = $((a * b))" ;;

4) echo "Result = $((a / b))" ;;

*)
```

Output:

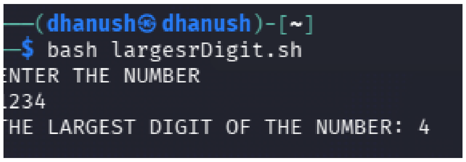


6.Largest Digit of Number

Program:

```
#!/bin/bash

echo "ENTER THE NUMBER"

read a

max=0

while [ $a -gt 0 ]

do

r=$((a % 10))

if [ $r -gt $max ]

then

max=$r

fi

a=$((a / 10))

done

echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

Output:



7. Palindrome String Check

Program:

```
#!/bin/bash

echo "ENTER THE STRING TO CHECK PALINDROME"

read str

len=$(echo -n "$str" | wc -c)

i=1

j=$((len / 2))

while [ $i -le $j ]

do

k=$(echo "$str" | cut -c $i)

l=$(echo "$str" | cut -c $len)

if [ "$k" != "$l" ]

then

echo "$str is not a palindrome"

exit

fi

i=$((i + 1))

len=$((len - 1))

done

echo "$str is a palindrome"
```

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ bash palindrome.sh
ENTER THE STRING TO CHECK PALINDROME
121
121 is a palindrome
```

8. Reverse of a Given Number

Program:

```
#!/bin/bash

echo "ENTER THE NUMBER"

read n

rnum=0

while [ $n -ne 0 ]

do

remainder=$((n % 10))

rnum=$((rnum * 10 + remainder))

n=$((n / 10))

done

echo "REVERSE OF THE NUMBER IS $rnum"
```

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ bash reverse.sh
ENTER THE NUMBER
12345
REVERSE OF THE NUMBER IS 54321
```